

Poornachandra G.

Agentic AI Engineer | LLMs, RAG & Multi-Agent Systems | AI/ML Solution Architecture

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PROFESSIONAL SUMMARY

AI/ML Engineer with 10+ years of experience designing and delivering end-to-end AI solutions, specializing in agentic AI, LLMs, RAG, and multi-agent systems. Experienced in solution architecture, agent orchestration, prompt engineering, retrieval pipelines, model evaluation, and responsible AI practices. Hands-on expertise building enterprise GenAI applications using LangChain, LangGraph, Azure AI Foundry, Azure AI Search, and MLflow. Skilled at translating complex business use cases into scalable, AI-driven automation that improves user experience, enhances process efficiency, and delivers production-ready solutions through software engineering best practices and cross-functional collaboration.

SKILLS & TECHNOLOGIES

Agentic AI & Orchestration	LangChain • LangGraph • CrewAI • OpenAI Agents SDK • Multi-Agent Systems • MCP (Model Context Protocol) • Agent Orchestration • Tool Integration • Memory Management • Human-in-the-Loop Workflows
RAG, Retrieval & Embeddings	RAG Pipeline Design • GraphRAG • Semantic Search • Azure AI Search • Neo4j • Embeddings • Chroma • Pinecone • Qdrant • Chunking Strategy Design • Retrieval Quality Validation • Golden Dataset Design
LLMs & Generative AI	OpenAI / Anthropic / Gemini APIs • Fine-Tuning • LoRA/QLoRA • Quantization • Ollama • AI Playgrounds
Prompt Engineering	Zero-shot / Few-shot / Role-based / Multi-turn Prompt Design • Prompt Lifecycle Management • Prompt Versioning • Context Engineering • Prompt Optimization
GenAI Evaluations & Responsible AI	Promptfoo • DeepEval • Ragas • LangSmith • Langfuse • LLM-as-a-Judge (LLMaaj) • Hallucination Analysis • Accuracy Benchmarking • Bias & Fairness Assessment • Red Teaming • Responsible AI Governance, Security & Privacy
Machine Learning	PyTorch • scikit-learn • LightGBM • XGBoost • Feature Engineering • SHAP • pandas • NumPy • SciPy • MLOps • MLflow
Cloud & Data Platforms	Azure (AI Foundry, AI Agents, AI Search, Blob Storage) • AWS (EC2, S3, Batch) • Databricks
GenAI Developer Tools	Claude Code • GitHub Copilot • Microsoft Copilot Studio • Cursor • Codex • Windsurf
Software Engineering & DevOps	Python • SQL • Bash • FastAPI • Docker • Git • GitHub Actions • CI/CD • Automated Testing • Pytest • Pydantic
Delivery & Leadership	Solution Architecture • Technical Roadmapping • Team Building & Mentoring • Agile/Scrum • JIRA • Confluence • AI enablement • Stakeholder Communication • HIPAA Compliance • Risk Management

PROFESSIONAL EXPERIENCE

Cellworks - Lead Clinical Translation

2024 - Present

- Designed multi-agent orchestration workflows for vendor order management, coordinating specialized agents, external tools, and contextual memory with human-in-the-loop validation at critical decision points.
- Built enterprise RAG pipelines with LangChain and Azure AI Search, deploying them on Azure AI Foundry to automate knowledge discovery across biomedical literature and proprietary knowledge bases.
- Assessed vector store solutions (Chroma, Pinecone, Qdrant) for RAG retrieval, tuning chunking strategy and embedding configuration to improve retrieval precision and reduce hallucination rates.
- Created prompt libraries and AI evaluation pipelines with Promptfoo and DeepEval, integrating version-controlled prompts and benchmarks into CI/CD to validate retrieval accuracy, consistency, hallucination, and regression before release.
- Developed golden datasets and benchmark queries to continuously evaluate retrieval quality and response accuracy, reducing post-deployment defects by 30%.
- Established responsible AI governance, security, and privacy checkpoints, including red-team testing and release gates, to guarantee safe and production-ready GenAI deployments.
- Led AI enablement programs and mentored engineers on agentic AI, prompt engineering, GenAI solutions, prompt engineering and evaluation best practices, driving org-wide GenAI adoption.

Cellworks - Senior Scientist, Reporting Manager

2021 - 2024

- Drove early adoption of LLM-assisted, RAG-based workflows for oncology therapy recommendations, integrating prompt engineering, knowledge indexing, and structured output evaluation.
- Designed an in-house ensemble-based ML variant reclassification pipeline, improving classification accuracy by 40%+; established reproducible ML workflows with MLflow for experiment tracking and model versioning.
- Built cloud-native data platforms on AWS and Databricks, engineering scalable ETL pipelines that feed ML and GenAI workloads.
- Built and led a technical team that conducted competitive technical analysis of market-leading healthcare AI/ML platforms while establishing core engineering standards, including modular design, automated CI/CD, and rigorous code reviews.
- Delivered reusable documentation, SOPs, and training modules to standardize process adoption across teams.

Cellworks - Senior Scientist, Clinical Translation

2018 - 2021

- Built a machine learning pipeline using longitudinal time-series data to predict early disease relapse, applying feature engineering and model validation methodology.
- Developed and benchmarked high-throughput data processing pipelines, applying structured accuracy, performance, and reproducibility evaluation methodology across pipeline versions.
- Created a validation framework for an edge-clinical case category, defining acceptance criteria, structured quality checkpoints, and classification guidelines to standardize simulation modeling.
- Built graph analytics pipelines with NetworkX to pinpoint influential nodes and derive network-based insights for data-driven decision making.
- Created reusable statistical modeling pipelines to enable data-driven discovery, employing reproducible experiment design and validation.

Cellworks - Scientist, Clinical Translation

2017 - 2018

- Engineered a Python rule-based variant classification framework to automate prioritization and standardize decision support.
- Developed an in-house variant annotation and filtering engine (Python, SQL), automating workflows and enabling downstream analytics.
- Acted as a technical bridge between engineering R&D and core development teams, accelerating product delivery through rapid prototyping and the automation of scalable data processing pipelines
- Implemented automated unit and integration tests with Pytest, increasing code reliability and enabling consistent validation of software releases through CI/CD.

Cellworks - Associate Scientist

2015 - 2017

- Automated report generation and ETL workflows using Python and SQL, reducing weekly processing time from ~15 hours to minutes.
- Standardized and automated large-scale data harmonization and ETL workflows across 500+ clinical cases and 10,000+ external records, enhancing data quality and processing consistency.
- Coordinated data standardization initiatives for data from multiple external partner institutions, ensuring consistent data formats, validation rules, and integration requirements.

INDEPENDENT PROJECTS

Inferverse: Created a unified interface for discovery, categorization, proxy configuration, and benchmarking of open generative AI APIs (NVIDIA NIM, Mistral, Groq, OpenRouter, Ollama, PublicAI)

PathoParse: LLM-assisted and rule-based pipelines to extract structured clinical metadata from unstructured pathology reports.

SQLChat: CrewAI multi-agent system translating natural-language queries into SQL, running statistical validation checks, and producing stakeholder-ready summaries.

EDUCATION

B.Tech, Biotechnology - JNTUA CEP

2015

PUBLICATIONS & ABSTRACTS

Co-author on 9 peer-reviewed publications and conference abstracts (*J Neurooncol*, *J Clin Oncol*, *Blood/ASCO*) on computational biomarker modeling and therapy-response prediction, 2018-2021. (Available on ORCID: [0009-0009-1610-1205](https://orcid.org/0009-0009-1610-1205))